Analysing Actor Networks in Quantum and Deep Tech Environments

Content Description

Emerging deep technologies such as quantum computing, fusion energy, and advanced robotics are characterised by long research and development cycles, high uncertainty, and strong interaction between multiple societal actors. Governments, research institutions, private companies, and civil society jointly shape the development of these technologies. Understanding how these actors interact and how innovation ecosystems evolve is therefore important for both policy makers and organisations operating in deep technology domains.

This interdisciplinary project aims to develop a data pipeline and analytical framework to identify, structure, and analyse actor networks in quantum and other deep technology fields. The project combines data engineering, computational text analysis, and innovation ecosystem research.

The first objective is to build a scalable data pipeline capable of collecting and structuring heterogeneous data sources related to deep technologies. Potential sources include national government strategies, policy reports, scientific publications, patents, news articles, social media posts, and company communications. Automated web scraping methods will be used to systematically collect relevant documents and store them in a structured database.

In a second step, the collected textual data will be processed using natural language processing (NLP) techniques. Named entity recognition will identify relevant actors such as universities, companies, research institutes, government bodies, and individuals. These actors will then be categorised according to a taxonomy inspired by the Quadruple Helix framework, distinguishing between academia, industry, government, and civil society. The project will also build upon an existing dataset of national quantum strategies in which actors are partially classified and annotated.

Following actor identification and classification, relation extraction and network analysis methods will be used to map connections between actors across documents and sources. This enables the construction of actor networks representing innovation ecosystems in quantum and other deep technology domains.

The resulting networks will be analysed to identify structural patterns, central actors, and emerging clusters. Comparisons across technological domains may reveal differences in how innovation ecosystems evolve. Additional techniques such as sentiment analysis or topic modelling may further help to understand how actors are positioned within technological narratives and policy discourse.

Relationship Between Project and Application Area

The project lies at the intersection of strategic management, innovation systems research, and computational data analysis. Emerging technologies such as quantum technologies involve complex interactions between multiple actors, including governments, research institutions, startups, and large corporations. Understanding these interactions is essential for analysing innovation ecosystems and the strategic positioning of organisations within them.

By analysing large collections of policy documents, media coverage, and other textual data, the project investigates how technological development is framed and coordinated across actors. Differences in actor networks across technological domains may provide insights into how innovation ecosystems evolve, how governments shape technology development through strategic initiatives, and how firms position themselves within these ecosystems.

From a strategic management perspective, analysing actor networks provides a better understanding of how organisations collaborate, compete, and build capabilities in emerging technology domains. The project therefore contributes to the study of technological innovation, organisational strategy, and ecosystem development in deep technology industries.

Accompanying Lectures

MGT001472 Applied Strategy and Organization: Strategies for International Corporations

This lecture examines the strategic challenges associated with internationalisation and digital transformation. Topics include corporate organisational structures, leadership, talent acquisition, knowledge management, and the impact of digitalisation on global operations. The course provides theoretical foundations that support the interpretation of the actor networks. Emerging technologies such as quantum technologies develop within complex innovation ecosystems involving governments, research institutions, startups, and multinational corporations. Understanding how organisations formulate strategies, build capabilities, and position themselves internationally helps explain the roles and interactions of different actors within these ecosystems, thereby strengthening the strategic interpretation of the empirical results.

WI001185 Strategic and International Management

This course provides a strong conceptual foundation in strategic decision-making, stakeholder management, and the formulation and implementation of strategies in complex environments. These perspectives are directly relevant to the proposed project, as actor networks in quantum and deep-tech ecosystems reflect how organizations position themselves strategically and form alliances. The course’s emphasis on international management is particularly valuable, given that quantum technology ecosystems are inherently global, involving cross-border collaborations between firms, research institutions, and governments. This knowledge will support the interpretation of network structures as outcomes of organizational strategies and competitive dynamics and strengthen the strategic analysis within the project.

Milestones

1. Literature Review and Conceptual Framework
    - Review of relevant literature on innovation ecosystems, deep technologies, and computational text analysis
    - Identification of suitable analytical frameworks (e.g. Quadruple Helix model)
2. Data Source Identification
    - Identification of relevant data sources such as government strategies, policy documents, scientific publications, news articles, and social media
3. Data Pipeline Development
    - Implementation of web scraping tools and database infrastructure
    - Automated data collection and storage
4. Data Processing and Actor Identification
    - Implementation of NLP techniques for entity extraction and classification
    - Structuring actors according to predefined taxonomies
5. Network Construction and Analysis
    - Identification of relationships between actors
    - Construction of actor networks and exploratory network analysis
6. Interpretation and Reporting
    - Comparison across technological domains
    - Integration of findings with innovation ecosystem theory

Schedule

Project duration: approximately three months

Month 1 - Conceptual Preparation and Data Collection

- Literature review and identification of relevant data sources
- Development of the data collection pipeline and database structure
- Initial data scraping and exploratory analysis

Month 2 - Data Processing and Model Development

- Data cleaning and preprocessing
- Implementation of entity extraction and actor classification methods
- Initial construction of actor networks

Month 3 - Network Analysis and Interpretation

- Network analysis and identification of structural patterns
- Comparative analysis across technological domains
- Integration of findings with theoretical frameworks and final report